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Interface for high-voltage applications

Abstract

An interface circuit, comprising: I/O circuitry configured to couple one or more first terminals with one or more second terminals; a voltage regulator configured to power the I/O circuitry; a restrictor element that is disposed between the voltage regulator and the I/O circuitry, the restrictor element being disposed on a circuit path that connects the voltage regulator to the I/O circuitry, the restrictor element being configured to open the circuit path and disconnect the voltage regulator from the I/O circuitry when an electrical current that is being supplied to the I/O circuitry exceeds a threshold; and a voltage suppressor that is disposed on the circuit path that connects the voltage regulator to the I/O circuitry, the voltage suppressor being disposed between the restrictor element and the I/O circuitry, the voltage suppressor being configured to limit, to a first voltage value, a voltage that is input into the I/O circuitry.

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Background/Summary

BACKGROUND

(1) An electronic system may include a controller as well as other control circuitry. The controller may communicate with an auxiliary device that has a different power supply than the controller. To ensure high reliability of the electronic system, the electronic system may utilize various protection mechanics that electrically separate the auxiliary device from the controller and prevent electrical overstress or failure events that take place at the auxiliary device from damaging the controller.

SUMMARY

(2) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

(3) According to aspects of the disclosure, an interface circuit is provided, comprising: an input-output (I/O) circuitry that is configured to couple one or more first terminals with one or more second terminals; a voltage regulator configured to power the I/O circuitry; a restrictor element that is disposed between the voltage regulator and the I/O circuitry, the restrictor element being disposed on a circuit path that connects the voltage regulator to the I/O circuitry, the restrictor element being configured to open the circuit path and disconnect the voltage regulator from the I/O circuitry when an electrical current that is being supplied to the I/O circuitry exceeds a threshold; a voltage suppressor that is disposed on the circuit path that connects the voltage regulator to the I/O

circuitry, the voltage suppressor being disposed between the restrictor element and the I/O circuitry, the voltage suppressor being configured to limit, to a first voltage value, a voltage that is input into the I/O circuitry; and a diagnostic circuit that is configured to: (i) compare a second voltage value to a voltage that is output by the voltage regulator, and (ii) when the voltage that is output by the voltage regulator matches the second voltage value, output an indication that the voltage regulator is malfunctioning, wherein the second voltage value is less than the first voltage value.

(4) According to aspects of the disclosure, an interface circuit is provided, comprising: a diagnostic terminal; an input-output (I/O) circuitry that is configured to couple one or more first terminals with one or more second terminals; a voltage regulator configured to power the I/O circuitry; and a diagnostic circuit that is configured to: (i) compare a first voltage value to a voltage that is output by the voltage regulator, and (ii) when the voltage that is output by the voltage regulator matches the first voltage value, output, on the diagnostic terminal, an indication that the voltage regulator is malfunctioning.

(5) According to aspects of the disclosure, an interface circuit is provided, comprising: an input-output (I/O) circuitry that is configured to couple one or more first terminals with one or more second terminals; a voltage regulator configured to power the I/O circuitry; a restrictor element that is disposed between the voltage regulator and the I/O circuitry, the restrictor element being disposed on a circuit path that connects the voltage regulator to the I/O circuitry, the restrictor element being configured to open the circuit path and disconnect the voltage regulator from the I/O circuitry when an electrical current that is being supplied to the I/O circuitry exceeds a threshold; and a voltage suppressor that is disposed on the circuit path that connects the voltage regulator to the I/O circuitry, the voltage suppressor being disposed between the restrictor element and the I/O circuitry, the voltage suppressor being configured to limit, to a first voltage value, a voltage that is input into the I/O circuitry.

(6) According to aspects of the disclosure, an interface circuit is provided, comprising: one or more first terminals for receiving at least one control signal from an electronic control unit (ECU); an input-output (I/O) circuit; a motor driver that is configured to drive a motor with an electrical current that is provided by a battery; a control logic that is configured to receive a control signal from the ECU and operate the motor driver based on the control signal, the control signal being received via the I/O circuit; and a first voltage regulator that is arranged to power the I/O circuit with the electrical current that is provided by the battery, the first voltage regulator being configured receive a same voltage as a voltage that is output by the battery and step down the received voltage.

(7) According to aspects of the disclosure, a system is provided, comprising: a battery; a motor; an ECU that is configured to a generate a control signal; an interface circuit that is configured to couple the motor to the ECU, the interface circuit including: an I/O circuit; a motor driver that is configured to drive the motor with an electrical current that is provided by the battery; a control logic that is configured to receive the control signal from the ECU, via the I/O circuit, and operate the motor driver based on the control signal; and a first voltage regulator that is arranged to power the I/O circuit with the electrical current that is provided by the battery, the first voltage regulator being configured receive a same voltage as a voltage that is output by the battery and step down the received voltage.

Description

BRIEF DESCRIPTION OF THE DRAWING FIGURES

(1) Other aspects, features, and advantages of the claimed invention will become more fully apparent from the following detailed description, the appended claims, and the accompanying drawings in which like reference numerals identify similar or identical elements. Reference

numerals that are introduced in the specification in association with a drawing figure may be repeated in one or more subsequent figures without additional description in the specification in order to provide context for other features.
(2) FIG. **1** is a diagram of an example of a system, according to the prior art.
(3) FIG. **2**A is a diagram of an example of a system, according to the aspects of the disclosure.
(4) FIG. **2**B is a diagram of an example of a system according to aspects of the disclosure;
(5) FIG. **3**A is a diagram of an example of a system according to aspects of the disclosure;
(6) FIG. **3**B is a diagram of an example of an I/O circuit, according to aspects of the disclosure;
(7) FIG. **3**C is a diagram of an example of a level translator, according to aspects of the disclosure; and
(8) FIG. **4** is a state diagram illustrating aspects of the operation of a system, according to aspects of the disclosure.

DETAILED DESCRIPTION

(9) In an embedded power system, a high-voltage microchip typically interfaces with a low-voltage microcontroller. This interface can be in the form of a bus across which digital input-output (I/O) is transferred. The internal power supply in the high-voltage component has to be regulated down to a suitable low voltage (VIO) that the microcontroller is rated for. The microcontroller will interface with a network of other peripheral components so it is imperative that even if one of those peripheral components develops a fault, the microcontroller is still able to control the others.
(10) In a high voltage system, there are multiple stress factors that can exacerbate latent defects in the device, such as reverse-battery, line transients, load dump, over-current due to short circuits, and over-voltage events. The regulator for VIO separates the high voltage from VIO by means of a pass device that it controls. In a safety-critical system, this pass device represents a potential single point of failure. If it fails, then VIO can exceed the Absolute Maximum Rating (AMR) of the connected microcontroller and cause the microcontroller to be rendered inoperable. A latent defect in the pass element that isn't detected in production could, during the service life of the system, cause a catastrophic failure.
(11) According to the present disclosure, an interface for coupling high voltage circuitry to a low-voltage microcontroller is provided. In one aspect, the interface may be arranged to employ a combination of detection and preventative elements to keep the microcontroller inside its AMR. In another aspect, the interface may be configured to prevent VIO from exceeding the AMR and sink current when the VIO exceeds a threshold. In yet another aspect, the interface may be configured to open the connection from VIO when excessive current flows.
(12) FIG. **1** is a diagram of an example of a system **100**, according to the prior art. As illustrated, the system **100** may include a controller **102** that is coupled to a high-voltage element **106** via an interface **104**. The controller **102** may include an Application-Specific Integrated Circuit (ASIC), a general-purpose processor (e.g., an ARM-based processor, MIPS processor, an x86 processor, etc.), a Field-Programmable Gate Array (FPGA), and or any other suitable type of processing circuitry. The high-voltage element **106** may include an industrial controller, a motor driver, a power converter and/or any other suitable type of circuitry or device.
(13) The controller **102** may be powered with a low-voltage signal (e.g., 5V). The low-voltage signal may be obtained by using a power manager **108** to step down a voltage provided by a battery **110**. The high-voltage element **106** may be powered directly by the battery **110**. In one example, the battery **110** may output 50V, and the power manager **108** may reduce the battery output to 5V.
(14) The interface **104** may be arranged to facilitate the flow of data or other signals between the high-voltage element **106** and the controller **102**. For example, in operation, the interface **104** may receive data or other signals from the controller **102** and forward the data or other signals to the high-voltage element **106**. As another example, the interface **104** may receive data or other signals from the high-voltage element **106** and forward the data or other signals to the controller **102**.
(15) In the example of FIG. **1**, the interface **104** is powered with a signal PWR that is provided by

the controller **102**. There are at least two disadvantages that are associated with this type of power layout. First, the number of terminals that are available on an integrated circuit is often limited, and the layout shown in FIG. **1** requires at least two terminals (Vcc and GND) on the controller **102** to be dedicated to powering the interface **104**, rather than being used for other purposes. Second, the layout shown in FIG. **1** limits the pool of controllers which the interface **104** can be used with since it requires controllers to have dedicated power supply pins for the interface **104**. In other words, the layout shown in FIG. **1** could make it impossible to use controllers that lack power supply pins for the interface **104** but are otherwise desirable to use with the interface **104** because they fit well a particular application.

(16) FIG. **2**A is a diagram of an example of a system **200**, according to aspects of the disclosure. The system **200** addresses at least some of the disadvantages of the power layout that is shown in FIG. **1**. Specifically, the system **200** features an interface **204** (between a controller **202** and a high-voltage element **206**) that is powered directly by a battery **210** (rather than by the controller **202**). A challenge posed by this design is that high voltage supplied directly to the interface **204**, by the battery **210**, may cause an overvoltage or overcurrent event at the I/O terminals of the controller **202**. The I/O terminals that are used to couple the controller **202** with the interface **204**. The overvoltage or overcurrent event may happen when the interface **204** (or voltage regulator inside the interface **204**) fails. The overvoltage or overcurrent event may cause a catastrophic failure of the controller **202**.

(17) According to aspects of the disclosure, the interface **204** is provided with several mechanisms that reduce the likelihood (or ideally prevent) the interface **204** from causing an overvoltage or overcurrent event at the I/O terminals of the controller **202**, while allowing the interface **204** to receive power from the battery **210** in a safe and dependable manner. In addition, the interface **204** is provided with a mechanism that generates a diagnostic signal that is configured to notify the controller **202** when the interface **204** is experiencing a failure. These mechanisms greatly improve the dependability of the interface **204** thus making it suitable for use in safety-critical applications. For instance, the mechanisms could help the interface **204** to achieve Automotive Safety Integrity Level D (ASIL D) certification, which may be necessary for the interface **204** to be used in many automotive applications. The mechanisms are discussed further below with respect to FIGS. **2**B-**4**.

(18) The controller **202** may include an Application-Specific Integrated Circuit (ASIC), a general-purpose processor (e.g., an ARM-based processor, MIPS processor, an x86 processor, etc.), a Field-Programmable Gate Array (FPGA), and or any other suitable type of processing circuitry. The high-voltage element **206** may include an industrial controller, a motor, a motor driver, an Application-Specific Integrated Circuit (ASIC), a general-purpose processor (e.g., an ARM-based processor, MIPS processor, an x86 processor, etc.), a Field-Programmable Gate Array (FPGA), and/or any other suitable type of circuitry, processing circuitry, or electrical device.

(19) The controller **202** may be powered with a low-voltage signal (e.g., 5V). The low-voltage signal may be obtained by using a power manager **208** to step down a voltage provided by a battery **210**. The high-voltage element **206** may be powered directly by the battery **210**. In one example, the battery **210** may output 50V, and the power manager **208** may reduce the battery output to 5V. However, the present disclosure is not limited to any specific configuration of the battery **210** or the power manager **208**.

(20) In some implementations, the interface **204** may be arranged to facilitate the flow of data or other signals between the high-voltage element **206** and the controller **202**. For example, in operation, the interface **204** may receive data or other signals from the controller **202** and forward the data or other signals to the high-voltage element **206**. As another example, the interface **204** may receive data or other signals from the high-voltage element **206** and forward the data or other signals to the controller **202**.

(21) Additionally or alternatively, in some implementations, the interface **204** may be configured to generate a diagnostic signal DIAG. The diagnostic signal DIAG may be generated under the

following circumstances: (i) when an internal voltage regulator of the interface **204** begins to fail, but before it has failed completely, and (ii) after the internal voltage regulator has failed. Under the nomenclature of the present disclosure, the voltage regulator is considered to have failed when the voltage output by the voltage regulator exceeds the AMR of the controller **202**. For example, when no abnormalities are present in the operation of the internal voltage regulator, the diagnostic signal DIAG may have a first value (e.g., '0'). On the other hand, when the internal voltage regulator begins to fail, or after it has failed, the diagnostic signal DIAG may assume a second value (e.g., '1'). The diagnostic signal may be provided to the controller **202**, and it may enable the controller **202** to take an action addressing the (impending) failure of the interface **204**.

(22) FIG. **2**B is a diagram of the system **200**, according to aspects of the disclosure. More specifically, FIG. **2**B shows one possible implementation of the interface **204** in further detail. As illustrated, the interface **204** may be coupled to the high-voltage element **206** via one or more data terminals T**2**. The interface circuit may include a voltage regulator **250**, a protection circuit **260**, an input-output (I/O) circuit **270**, and a voltage comparator **280**.

(23) The voltage regulator **250** may be coupled to the battery **210** via a power terminal PWR of the interface **204**. The voltage regulator may receive a voltage VBB from the battery **210** and step down the voltage VBB to a voltage VIO. In the example of FIG. **2**B, the voltage (relative to ground) at node N**1** is VBB and the voltage (relative to ground) at node N**2** is VIO. In one example, VBB may equal 85V and VIO may equal 5 or 3.3V. The voltage regulator may include a MOSFET **254**, a voltage subtractor **252**, a non-linear regulator and/or any other suitable type of component or circuit. The drain of MOSFET **254** may be coupled to node N**1**, the source of MOSFET **254** may be coupled to node N**2**, and the gain of the MOSFET **254** may be coupled to the output of the voltage subtractor **252**. The voltage subtractor **252** may be configured to subtract the voltage at the source of the MOSFET **254** from a reference voltage VREF. The reference voltage VREF may be equal to the value that VIO is desired to have (e.g., 5V or 3.3V) during normal operation of the voltage regulator **250** (or it may be a lower value and a voltage divider may be used in the feedback path). Although in the present example the voltage regulator **250** is implemented as a linear regulator, it will be understood that the present disclosure is not limited to any specific implementation of the voltage regulator **250**.

(24) The protection circuit **260** may be coupled to the voltage regulator **250** at node N**2** and to the I/O circuit **270** at node N**3**. In other words, the protection circuit **260** may be disposed on the electrical path between nodes N**2** and N**3**. The I/O circuit **270** may be powered by an electrical current that is received through the electrical path between nodes N**2** and N**3**. The protection circuit **260** may be configured to open (or otherwise interrupt) the path between nodes N**2** and N**3** in the event of an overcurrent or overvoltage conduction. The protection circuit **260** may fulfill the following functions: (i) protect the I/O circuit **270** from damage by preventing excessive current or voltage from reaching the I/O circuit **270**, (ii) prevent the controller **202** from damage by preventing excessive voltage or current from being applied at terminals T**1**. The protection circuit may include a fuse **263** that is coupled to a Zener diode **256**, as shown. The Zener diode **256** may be configured to clamp the value of VIO at a clamp voltage VCL (e.g., 6.5V), and the fuse **263** may be configured to interrupt the flow of electrical current through the path between nodes N**2** and N**3** when it exceeds the maximum short circuit current IIO for the I/O circuit **270**. In operation, when the voltage at the Zener diode **256** surpasses the value of VCL, the Zener diode **256** may increase the electrical current through the fuse **263**, thus causing the fuse **263** to open the path between nodes N**2** and N**3** and interrupt the flow of current to the I/O circuit **270** in this way. The fuse **263** may open the path between nodes N**2** and N**3** when the electrical current through the fuse **263** exceeds the value of the maximum restrictor current ICN.

(25) In some implementations, for the early warning to be effective, the fault reaction time of the system **200** needs to be designed such that when the signal DIAG is asserted, it can be acted upon by the controller **202** to isolate the fault and avoid the protection circuit **260** from having to

activate. In the event of a fault causing VIO to rise to the battery voltage VBB, the signal DIAG should assert and be acted upon in microseconds. The clamp circuit (e.g., the Zener diode **256**) may protect the I/O circuit **270** during the fault reaction time. If the maximum fault reaction time is exceeded, the fuse blow time (of the fuse **263**) should be designed to be shorter than the time the I/O circuit **270** can endure the excess power being dissipated in its components. This allows time for the system **200** to be put in a safe state before the fuse **263** opens the line between nodes N**2** and N**3**, and communication with the controller **202** is lost.
(26) Although in the example of FIG. **2**B a Zener diode is used to clamp the voltage at node N**3**, it will be understood that alternative implementations are possible in which another type of clamp circuit is used, such as a resistor. Although in the example of FIG. **2**B, the fuse **263** is used to constrict the current through the path N**2**-N**3**, alternative implementations are possible in which another current restrictor is used, such as an active current limiter.
(27) The I/O circuit **270** may include any suitable type of circuitry that is configured to (i) receive a first data set at terminals T**1** and output a second dataset at terminals T**2**. In some instances, the I/O circuit **270** may be implemented as discussed further below with respect to FIGS. **3**C-D.
(28) The voltage comparator **280** may be configured to compare the voltage at node N**2** (i.e., the output voltage of the voltage regulator **250** to a voltage value VWARN and output a signal DIAG based on an outcome of the comparison. For example, the diagnostic signal DIAG may be set to a first value (e.g., ‘0’) when VIO is less than VWARN, and the diagnostic signal DIAG may be set to a second value (e.g., ‘1’) when VIO is greater than or equal to VWARN. Under the nomenclature of the present disclosure, the voltage comparator **280** is also referred to as a diagnostic circuit.
(29) In some implementations, the value of VWARN may be less than the value of the clamp voltage VCL of the Zener diode **256**—for example, VCL may be equal to 6.5, VWARN may be equal to 5.7V, and the value which VIO is expected to have absent a failure of the voltage regulator **250** may be 5V. In some implementations, the value of VCL may be less than or equal to the value of the AMR for the controller **202**. Additionally or alternatively, in some implementations, the value of VCL may be less than or equal to the AMR of the ECU **301** (shown in FIG. **3**). By way of example, in some implementations, VCL may be equal to 85% of the AMR for the controller **202** (or the ECU **301**).
(30) The diagnostic signal DIAG may provide a notice when the value of VIO begins to exceed its normal bounds but has not yet crossed the critical threshold represented by VCL. In this way, the controller **202** (or the high-voltage element **206**) may take a responsive action after the value of VIO exceeds VWARN, but before the data connection between the controller **202** and the high-voltage element is severed (as a result of the protection circuit reaching the clamp voltage VCL of the Zener diode **256**).
(31) FIG. **3**A is a diagram of an example of a system **300**. The system **300**, according to the present example is an electric vehicle. However, alternative implementations are possible in which the system **300** includes any suitable type of electromechanical system. In some respects, FIG. **3**A is provided to illustrate another possible implementation of the interface **204**.
(32) As illustrated, the system **300** may include the battery **210**, an electronic control unit (ECU) **301**, the interface **204**, and an external bridge and motor **310** (hereinafter “motor **310**”). The ECU **301** may be coupled to the motor **310** via the interface **204**. The ECU **301** may include any suitable type of electronic control unit that is configured to control one or more characteristics of the operation of the motor **310**, such as speed or torque, for example. The motor **310** may include any suitable type of electric motor, such as a 3-phase motor, a 5-phase motor, a 7-phase motor, etc.
(33) According to the present example, the motor **310** is a wheel motor and the ECU **301** is the main computer of an electric vehicle. However, as noted above, the present disclosure is not limited to any specific implementation of the system **300**. In this regard, it will be understood that the system **300** may be any suitable type of machinery that includes an electric motor. Furthermore, it will be understood that the ECU **301** may include any suitable type of controller, such as an

industrial controller for example. Although in the example of FIG. **3**A, the high voltage source used to power the interface **204** is a battery, it will be understood that alternative implementations are possible in which another high-voltage source is used instead.
(34) In the example of FIG. **3**A, the interface **204** is a motor control interface, and it includes a driving logic **306** and a motor driver **308**, in addition to the voltage regulator **250**, the voltage regulator **262**, the protection circuit **260**, the I/O circuit **270**, and the voltage comparator **280**. The configuration of the voltage comparator **280** and the voltage regulator **250** is discussed above with respect to FIG. **2**B. The configuration of the voltage regulator **262** may be the same or similar to the voltage regulator **250**. However, the present disclosure is not limited to any specific configuration of the voltage regulator **262**.
(35) The protection circuit **260** may be disposed on the circuit path **312** that couples the voltage regulator **250** with the I/O circuit **270**. As discussed above with respect to FIG. **2**B, the protection circuit **260** may be configured to open the circuit path **312** and disconnect the voltage regulator **250** from the I/O circuit in the event of a malfunction of the voltage regulator **250**. As noted above, the protection circuit **260** may open the circuit path **312** when VIO (i.e., the voltage output by the voltage regulator **250**) exceeds the breakdown voltage VCL of the Zener diode **256** or when electrical current through the protection circuit **260** exceeds the maximum restrictor current ICN of the fuse **263**.
(36) As discussed above with respect to FIG. **2**B, the voltage comparator **280** may be configured to compare the value of voltage VIO to the value VWARN, and set the signal DIAG based on an outcome of the comparison. In some implementations, the diagnostic signal may be output on a fault terminal FT of the interface **204**. The fault terminal FT may be coupled to a GPIO pin of the ECU **301**, a direct memory access (DMA) interface of the ECU **301**, and/or an interrupt controller of the ECU **301**.
(37) The driving logic **306** may include any suitable type of processing circuitry, such as an application-specific processing circuitry, or a general-purpose processing circuitry, for example. The driving logic **306** may be configured to receive data over the I/O circuit **270**. The data received over the I/O circuit **270** may include one or more digital control signals that are generated by the ECU **301**. The digital control may specify one or more characteristics of the operation of the motor **310**, such as torque or speed for example. The driving logic **306** may be configured to generate one or more control signals **326** and provide the control signals **326** to the motor driver **308**. The control signals **326** may be generated based on the data that is received over the I/O circuit **270**. The control signals **326** may encode at least some of the information that is received over the I/O circuit **270**. For example, if the data specifies a torque value, the control signals **326** may encode the torque value; if the data specifies a speed value, the control signals **326** may encode the speed value, etc. In addition, the driving logic **306** may output feedback to control signals provided by the ECU **301** via the I/O circuit **270**. Throughout the disclosure, the term "logic" as used in the phrase "driving logic" shall refer to electronic circuitry, alone or in combination with software, that is configured to perform one or more operations.
(38) The motor driver **308** may include any suitable type of electronic circuitry for driving the motor **310**. The motor driver **308** may generate a PWM signal **328** (or another power signal) based on the control signal **326**. The PWM signal **328** may be subsequently used to power the external motor and bridge. The PWM signal may be generated by modulating the current that is output by the battery **210**. The PWM signal **328** may be generated so as to cause the motor **310** to operate in accordance with one or more operational parameters that are specified by the control signal **326**.
(39) According to the example of FIG. **3**A, the voltage regulator **262** is arranged to step down the voltage VBB, which is provided by the battery **210**, to a voltage VDD. According to the present example, VDD is equal to 5V. However, the present disclosure is not limited to any specific value of VDD.
(40) The voltage comparator **280** may be arranged to receive power from the voltage regulator **262**

via a circuit path **322**. A protection circuit **264** may be optionally disposed on the circuit path **322**, as shown. The protection circuit **264** may have the same or similar configuration as the protection circuit **260**. The protection circuit **264** may be configured to open the circuit path **322** (and disconnect the voltage comparator **280** from the voltage regulator **262**), when the voltage VDD that is output by the voltage regulator **262** exceeds a voltage threshold or when the current through the protection circuit **264** exceeds a current threshold.
(41) The driving logic **306** may be arranged to receive power from the voltage regulator **262** via a circuit path **332**. A protection circuit **265** may be optionally disposed on the circuit path **332**, as shown. The protection circuit **265** may have the same or similar configuration as the protection circuit **260**. The protection circuit **265** may be configured to open the circuit path **332** (and disconnect the driving logic **306** from the voltage regulator **262**), when the voltage VDD that is output by the voltage regulator **262** exceeds a voltage threshold or when the current through the protection circuit **265** exceeds a current threshold.
(42) The motor driver **308** may be arranged to receive power from the voltage regulator **262** via a circuit path **342**. A protection circuit **266** may be optionally disposed on the circuit path **342**, as shown. The protection circuit **266** may have the same or similar configuration as the protection circuit **260**. The protection circuit **266** may be configured to open the circuit path **342** (and disconnect the motor driver **308** from the voltage regulator **262**), when the voltage VDD that is output by the voltage regulator **262** exceeds a voltage threshold or when the current through the protection circuit **266** exceeds a current threshold.
(43) In some instances, a respective protection circuit **267** may be disposed between each of the terminals T**1** and the I/O circuit **270**. Each protection circuit **267** may have the same or similar configuration as the protection circuit **260**. Each protection circuit **267** may be configured to open the line that connects the protection circuit's **267** respective terminal T**1** to the I/O circuit **270**, when the voltage across the line exceeds a voltage threshold or when the current through the line exceeds a current threshold.
(44) Any two (or all) of the protection circuits **264-267** may have the same or different voltage thresholds. Any two (or all) of the protection circuits **264-267** may have the same or different current thresholds. In some implementations, the voltage threshold of any of the protection circuits **264-267** may be less than or equal to the clamp voltage of a clamp circuit that is part of that protection circuit. In some implementations, the current threshold of any of the protection circuits **264-267** may be less than or equal to the maximum restrictor current of a current restrictor that is part of that protection circuit.
(45) FIG. **3**B is a diagram of an example of the I/O circuit **270** in accordance with one particular implementation. As illustrated, the I/O circuit may include a plurality of lines **361**. Each of lines **361** may include a respective I/O driver **362** that is coupled in series with a respective level translator **363**. Each of the I/O drivers **362** may include either an input driver or an output driver.
(46) FIG. **3**C shows an example of a level translator **363**, according to aspects of the disclosure. The level translator **363** that is shown in FIG. **3**C may be the same or similar to any of the level translators **363** that are shown in FIG. **3**B. As illustrated, the level translator **363** may include a terminal **371** that is coupled to the input of a buffer **372** (e.g., see FIG. **3**A). The buffer **372** may be powered with the voltage regulator **250**, and arranged to receive the voltage VIO from the voltage regulator **250**. The level translator **361** may include a load **374** that is coupled to the emitter of a transistor **375**, as shown. The emitter of transistor **375** may also be coupled to the input of an inverter **366**. The collector of transistor **375** may be coupled to ground. The inverter **366** may be powered with the voltage regulator **262** and arranged to receive the voltage VDD from the voltage regulator **262**. The output of the inverter **374** may be coupled to a terminal **376**. The terminal **376** may be coupled to the driving logic **306**.
(47) A dielectric barrier **373** may be disposed between the buffer **372** and the base of transistor **475**. Specifically, the output of buffer **372** may be coupled to the dielectric barrier **373**, and the base of

transistor **375** may be coupled to the dielectric barrier **373**. The dielectric barrier **373** may prevent current from travelling between the transistor **375** and the buffer **375**. The dielectric barrier **373** may be configured to separate the voltage domain of the voltage regulator **250** from the voltage domain of the voltage regulator **262**. Depending on the application and technology, the dielectric barrier **373** may be simply a MOS or other insulated-gate transistor with suitable breakdown voltage or another isolated structure.

(48) According to the example of FIG. **3**C, each of lines **361** is configured to couple a different one of the terminals T**1** to a corresponding terminal of the driving logic **306**. However, in the implementation discussed with respect to FIG. **2**D, each of the lines **361** may be configured to couple a different one of the terminals T**1** to a corresponding one of the terminals T**2**. In this implementation, the dielectric barrier may be configured to separate the voltage domain of the voltage regulator **250** from a voltage domain that is associated with the terminals T**2**. The voltage domain associated with the terminals T**2** may be a voltage domain of a voltage regulator that is configured to step down the voltage VBB and power one or more components of the voltage element **206** (such as a control circuitry of the element **206**).

(49) FIG. **4** is a state diagram of illustrating aspects of the operation of the ECU **301**, according to aspects of the disclosure. As illustrated, the ECU **301** may be in a normal operating state **402**, in a heightened alert state **404**, and in a disconnected state **406**. It will be understood that FIG. **4** is provided for illustrative purposes only, and that the ECU **301** may have many other states in addition to those that are shown in FIG. **4**.

(50) When the ECU **301** is in the state **402**, the ECU **301** may be able to send signals to the driving logic **306** and/or receive signals from the driving logic **306**. In general, when the ECU **301** is in the state **402**, the voltage VIO that is output by the voltage regulator **250** may be within normal operating bounds (e.g., less than the value VWARN).

(51) When the ECU **301** is in the state **404**, VIO may be greater than VWARN, but less than VCL. In other words, when the ECU **301** is in the state **404**, the ECU **301** may be still in possession of the ability to send and receive signals from the driving logic **306** of the interface **204**, but it may also be on the way of losing this ability. While the ECU **301** is in the state **404**, the ECU **301** may perform actions that are aimed at" (i) preparing the system **300** for the eventuality of the communication between the ECU **301** and the driving logic **306** being severed, and/or (ii) returning the system to a state in which the value of VIO is less than VWARN and VCL. For example, in some implementations, when the system **300** is in the state **404**, the system **300** may cause the driving logic **306** to begin operating autonomously of the ECU **301** or turn off the motor **310**. Additionally or alternatively, as noted above, when the ECU **301** is in the state **404**, the ECU **301** may take a corrective action aimed at returning the ECU **301** to a state in which the value of VIO is less than VWARN an VCL. For example, the corrective action may include switching the I/O circuit **270** to an alternative power supply. The alternative power supply may be another voltage regulator that is provided in the interface **204** as a backup for voltage regulator **250**.

(52) When the ECU **301** is in the state **406**, the circuit path **322** may be open and the I/O circuit **270** may be turned off. As a result, the ECU **301** may be disconnected from the driving logic **306** and/or the motor driver **308** of the interface **204**.

(53) In one example, the ECU **301** may transition from state **402** to state **404** when the diagnostic signal DIAG is set to logic-high. The ECU **301** may transition from state **404** to state **406** when connection is lost with the driving logic **306** of the interface **204**. The ECU **301** may transition from state **402** directly to state **406** when connection is lost with the driving logic **306** of the interface **204**. The ECU **301** may return from state **404** to state **402** when the value of the diagnostic signal DIAG is set back to logic-low (e.g., as a result of the value of the voltage supplied to the I/O circuit **270** falling below the value of VWARN).

(54) As discussed above with respect to FIGS. **2**B and **3**, the value of VWARN may be less than the clamp voltage VCL of the protection circuit **260**, and the clamp voltage VCL may be less than or

equal to the AMR of the ECU **301**. The diagnostic signal DIAG may be set to a logic-high in response to the voltage VIO that is output by the voltage regulator **250** exceeding VWARN, and it may return to logic-low in response to VIO falling below VIO. The I/O circuit **270** may be turned off when the VIO exceeds VCL and/or when the current through the protection circuit **260** exceeds the maximum restrictor current ICN of the protection circuit **260**. The interface circuit **204** may be implemented as an integrated circuit and packaged by using any suitable type of semiconductor packaging technology.
(55) As used in this application, the word "exemplary" is used herein to mean serving as an example, instance, or illustration. Any aspect or design described herein as "exemplary" is not necessarily to be construed as preferred or advantageous over other aspects or designs. Rather, use of the word exemplary is intended to present concepts in a concrete fashion. As used throughout the disclosure, the term product may include a physical object that is being bought and sold, a service, and/or anything else that can be purchased and solved. FIGS. **2**A-**4** are provided as an example only.
(56) Additionally, the term "or" is intended to mean an inclusive "or" rather than an exclusive "or". That is, unless specified otherwise, or clear from context, "X employs A or B" is intended to mean any of the natural inclusive permutations. That is, if X employs A; X employs B; or X employs both A and B, then "X employs A or B" is satisfied under any of the foregoing instances. In addition, the articles "a" and "an" as used in this application and the appended claims should generally be construed to mean "one or more" unless specified otherwise or clear from context to be directed to a singular form.
(57) To the extent directional terms are used in the specification and claims (e.g., upper, lower, parallel, perpendicular, etc.), these terms are merely intended to assist in describing and claiming the invention and are not intended to limit the claims in any way. Such terms, do not require exactness (e.g., exact perpendicularity or exact parallelism, etc.), but instead it is intended that normal tolerances and ranges apply. Similarly, unless explicitly stated otherwise, each numerical value and range should be interpreted as being approximate as if the word "about", "substantially" or "approximately" preceded the value of the value or range.
(58) Moreover, the terms "system," "component," "module," "interface,", "model" or the like are generally intended to refer to a computer-related entity, either hardware, a combination of hardware and software, software, or software in execution. For example, a component may be, but is not limited to being, a process running on a processor, a processor, an object, an executable, a thread of execution, a program, and/or a computer. By way of illustration, both an application running on a controller and the controller can be a component. One or more components may reside within a process and/or thread of execution and a component may be localized on one computer and/or distributed between two or more computers.
(59) Although the subject matter described herein may be described in the context of illustrative implementations to process one or more computing application features/operations for a computing application having user-interactive components the subject matter is not limited to these particular embodiments. Rather, the techniques described herein can be applied to any suitable type of user-interactive component execution management methods, systems, platforms, and/or apparatus.
(60) While the exemplary embodiments have been described with respect to processes of circuits, including possible implementation as a single integrated circuit, a multi-chip module, a single card, or a multi-card circuit pack, the described embodiments are not so limited. As would be apparent to one skilled in the art, various functions of circuit elements may also be implemented as processing blocks in a software program. Such software may be employed in, for example, a digital signal processor, micro-controller, or general-purpose computer.
(61) Some embodiments might be implemented in the form of methods and apparatuses for practicing those methods. Described embodiments might also be implemented in the form of program code embodied in tangible media, such as magnetic recording media, optical recording

media, solid state memory, floppy diskettes, CD-ROMs, hard drives, or any other machine-readable storage medium, wherein, when the program code is loaded into and executed by a machine, such as a computer, the machine becomes an apparatus for practicing the claimed invention. Described embodiments might also be implemented in the form of program code, for example, whether stored in a storage medium, loaded into and/or executed by a machine, or transmitted over some transmission medium or carrier, such as over electrical wiring or cabling, through fiber optics, or via electromagnetic radiation, wherein, when the program code is loaded into and executed by a machine, such as a computer, the machine becomes an apparatus for practicing the claimed invention. When implemented on a general-purpose processor, the program code segments combine with the processor to provide a unique device that operates analogously to specific logic circuits. Described embodiments might also be implemented in the form of a bitstream or other sequence of signal values electrically or optically transmitted through a medium, stored magnetic-field variations in a magnetic recording medium, etc., generated using a method and/or an apparatus of the claimed invention.

(62) It should be understood that the steps of the exemplary methods set forth herein are not necessarily required to be performed in the order described, and the order of the steps of such methods should be understood to be merely exemplary. Likewise, additional steps may be included in such methods, and certain steps may be omitted or combined, in methods consistent with various embodiments.

(63) Also, for purposes of this description, the terms “couple,” “coupling,” “coupled,” “connect,” “connecting,” or “connected” refer to any manner known in the art or later developed in which energy is allowed to be transferred between two or more elements, and the interposition of one or more additional elements is contemplated, although not required. Conversely, the terms “directly coupled,” “directly connected,” etc., imply the absence of such additional elements.

(64) As used herein in reference to an element and a standard, the term “compatible” means that the element communicates with other elements in a manner wholly or partially specified by the standard, and would be recognized by other elements as sufficiently capable of communicating with the other elements in the manner specified by the standard. The compatible element does not need to operate internally in a manner specified by the standard.

(65) It will be further understood that various changes in the details, materials, and arrangements of the parts which have been described and illustrated in order to explain the nature of the claimed invention might be made by those skilled in the art without departing from the scope of the following claims.

Claims

1. An interface circuit, comprising: an input-output (I/O) circuitry that is configured to couple one or more first terminals with one or more second terminals, the I/O circuitry including circuitry to receive a data set at a plurality first terminals of the interface circuit and output the data set on a plurality of second terminals of the interface circuit, the I/O circuitry including a plurality of level translators, each level translator being coupled between a different respective one of the first terminals and a different respective one of the second terminals; a voltage regulator configured to power the I/O circuitry; a fuse that is disposed between the voltage regulator and the I/O circuitry, the fuse being disposed on a circuit path that connects the voltage regulator to the I/O circuitry, the fuse being configured to open the circuit path and disconnect the voltage regulator from the I/O circuitry when an electrical current that is being supplied to the I/O circuitry exceeds a threshold; a Zener diode that is coupled to the circuit path that connects the voltage regulator to the I/O circuitry, the Zener diode being configured to increase the electrical current that is being supplied to the I/O circuitry and cause the fuse to disconnect the I/O circuitry from the voltage regulator when the voltage that is input into the I/O circuitry exceeds a clamping voltage of the Zener diode;

and a diagnostic circuit that is configured to: (i) compare a reference voltage to a voltage that is output by the voltage regulator, and (ii) when the voltage that is output by the voltage regulator matches the reference voltage value, output an indication that the voltage regulator is malfunctioning, wherein the reference voltage is less than the clamping voltage of the Zener diode.

2. The interface circuit of claim 1, wherein the I/O circuitry includes at least one I/O driver that is coupled in series with a level translator.

3. The interface circuit of claim 1, wherein the diagnostic circuit is powered by a voltage source that is disposed outside of the circuit path that connects the voltage regulator to the I/O circuitry.

4. The interface circuit of claim 1, wherein each of the level translators is configured to provide a dielectric barrier that is arranged to separate a voltage domain of the voltage regulator from a voltage domain that is associated with the second terminal that is coupled to the level translator.

5. An interface circuit, comprising: a diagnostic terminal; an input-output (I/O) circuitry that is configured to couple one or more first terminals with one or more second terminals, the I/O circuitry including circuitry to receive a data set at a plurality first terminals of the interface circuit and output the data set on a plurality of second terminals of the interface circuit, the I/O circuitry including a plurality of level translators, each level translator being coupled between a different respective one of the first terminals and a different respective one of the plurality of second terminals; a voltage regulator configured to power the I/O circuitry; and a restrictor element configured to disconnect the voltage regulator from the I/O circuitry when an overvoltage event occurs in the interface circuit; a diagnostic circuit that is configured to: (i) compare a first voltage value to a voltage that is output by the voltage regulator, and (ii) when the voltage that is output by the voltage regulator matches the first voltage value, output, on the diagnostic terminal, an indication that the interface circuit is in a state in which there is an increased likelihood for the I/O circuitry to be turned off as a result of the overvoltage event occurring, the overvoltage event occurring when the voltage supplied by the voltage regulator crosses a second voltage value, the second voltage value being greater than the first voltage value.

6. The interface circuit of claim 5, wherein the diagnostic circuit is powered by a voltage source that is disposed outside of a circuit path that connects the voltage regulator to the I/O circuitry.

7. The interface circuit of claim 5, wherein the restrictor element includes a fuse.

8. The interface circuit of claim 5, wherein each of the level translators is configured to provide a dielectric barrier that is arranged to separate a voltage domain associated with the first terminal that is coupled to the level translator from a voltage domain that is associated with the second terminal is coupled to the level translator.

9. The interface circuit of claim 5, further comprising a voltage suppressor wherein the voltage suppressor is configured to increase an electrical current that is being supplied to the I/O circuitry and cause the restrictor element to disconnect the I/O circuitry from the voltage regulator when the voltage that is output by the voltage regulator has exceeded the first voltage value by a predetermined amount.

10. The interface circuit of claim 9, wherein the voltage suppressor includes a Zener diode.

11. The interface circuit of claim 9, wherein the restrictor element includes a fuse, and the voltage suppressor includes a clamp circuit.

12. An interface circuit, comprising: an input-output (I/O) circuitry that is configured to couple one or more first terminals with one or more second terminals, the I/O circuitry including circuitry to receive a data set at a plurality first terminals of the interface circuit and output the data set on a plurality of second terminals of the interface circuit, the I/O circuitry including a plurality of level translators, each level translator being coupled between a different respective one of the first terminals and a different respective one of the plurality of second terminals; a voltage regulator configured to power the I/O circuitry; a restrictor element that is disposed between the voltage regulator and the I/O circuitry, the restrictor element being disposed on a circuit path that connects the voltage regulator to the I/O circuitry, the restrictor element being configured to open the circuit

path and disconnect the voltage regulator from the I/O circuitry when an electrical current that is being supplied to the I/O circuitry exceeds a first threshold; and a voltage suppressor that is coupled to the circuit path that connects the voltage regulator to the I/O circuitry, the voltage suppressor being configured to limit a voltage that is input into the I/O circuitry; and a diagnostic circuit that is configured to: (i) compare a second threshold to a voltage that is output by the voltage regulator, and (ii) when the voltage that is output by the voltage regulator matches the second threshold, output an indication that the voltage regulator is malfunctioning, wherein the second threshold is less than the first threshold.

13. The interface circuit of claim 12, wherein the voltage suppressor is configured to increase the electrical current that is being supplied to the I/O circuitry and cause the restrictor element to disconnect the I/O circuitry from the voltage regulator when the voltage that is input into the I/O circuitry is equal to the first threshold or has exceeded the first threshold by a predetermined amount.

14. The interface circuit of claim 12, wherein each of the level translators is configured to provide a dielectric barrier that is arranged to separate a voltage domain of the voltage regulator from a voltage domain that is associated with the second terminal that is coupled to the level translator.

15. The interface circuit of claim 12, wherein the restrictor element includes a fuse, and the voltage suppressor includes a clamp circuit.

16. An interface circuit, comprising: an input-output (I/O) circuit; a motor driver that is configured to drive a motor with an electrical current that is provided by a battery; a control logic that is configured to receive a control signal from an electronic control unit (ECU) and operate the motor driver based on the control signal, the control signal being received via the I/O circuit, the I/O circuit including circuitry to receive a control signal at a plurality of input terminals and output the control signal to the control logic, the I/O circuit including a plurality of level translators, each level translator being coupled between a different respective one of the input terminals and the control logic; and a first voltage regulator that is arranged to power the I/O circuit with the electrical current that is provided by the battery, the first voltage regulator being configured receive a same voltage as a voltage that is output by the battery and step down the received voltage; a restrictor element configured to disconnect the first voltage regulator from the I/O circuit when an overvoltage event occurs in the interface circuit; a diagnostic circuit that is configured to: (i) compare a first voltage value to a voltage that is output by the first voltage regulator, and (ii) when the voltage that is output by the first voltage regulator matches the first voltage value, output, on a diagnostic terminal, an indication that the interface circuit is in a state in which there is an increased likelihood for the I/O circuit to be turned off, the overvoltage event occurring when the voltage supplied by the first voltage regulator crosses a second voltage value, the second voltage value being greater than the first voltage value.

17. The interface circuit of claim 16, wherein the restrictor element includes a fuse.

18. The interface circuit of claim 16, further comprising a voltage suppressor that is coupled to a circuit path that connects the first voltage regulator to the I/O circuit, the voltage suppressor being configured to cause the diagnostic circuit to disconnect the first voltage regulator from the I/O circuit when the voltage that is being supplied by the first voltage regulator to the I/O circuit exceeds the first voltage.

19. The interface circuit of claim 18, wherein the diagnostic circuit is powered by a second voltage regulator that is disposed outside of the circuit path that connects the first voltage regulator to the I/O circuit, the second voltage value being less than the first voltage value.

20. The interface circuit of claim 16, wherein each of the level translators is configured to provide a dielectric barrier.

21. A system, comprising: a battery; a motor; an ECU that is configured to a generate a control signal; an interface circuit that is configured to couple the motor to the ECU, the interface circuit including: an I/O circuit; a motor driver that is configured to drive the motor with an electrical

current that is provided by the battery; a control logic that is configured to receive the control signal from the ECU, via the I/O circuit, and operate the motor driver based on the control signal, the I/O circuit including circuitry to receive the control signal at a plurality of input terminals and output the control signal to the control logic, the I/O circuit including a plurality of level translators, each level translator being coupled between a different respective one of the input terminals and the control logic; and a first voltage regulator that is arranged to power the I/O circuit with the electrical current that is provided by the battery, the first voltage regulator being configured receive a same voltage as a voltage that is output by the battery and step down the received voltage; and a restrictor element being configured to disconnect the first voltage regulator from the I/O circuit when a voltage that is being supplied by the first voltage regulator the I/O circuit exceeds a first threshold, and a diagnostic circuit that is configured to: (i) compare a second threshold to a voltage that is output by the first voltage regulator, and (ii) when voltage that is output by the first voltage regulator matches the second threshold, output, the ECU, an indication that the first voltage regulator is malfunctioning, wherein the second threshold is less than the first threshold.

22. The system of claim 21, wherein the restrictor element includes a fuse.

23. The system of claim 21, wherein the interface circuit further includes a voltage suppressor that is coupled to a circuit path that connects the first voltage regulator to the I/O circuit, the voltage suppressor being configured to cause the diagnostic circuit to disconnect the first voltage regulator from the I/O circuit when the voltage that is being supplied by the first voltage regulator to the I/O circuit exceeds the first threshold.

24. The system of claim 23, wherein the diagnostic circuit is powered by a second voltage regulator that is disposed outside of the circuit path that connects the first voltage regulator to the I/O circuit.

25. The system of claim 21, wherein each of the level translators is configured to provide a dielectric barrier.